ESPI



CONSTELLATING COMPUTE

CHINA'S DOMESTIC POLICY TOWARDS ORBITAL DATA CENTRE LEADERSHIP

EXECUTIVE SUMMARY

China is one of the world's key players in the development of new advanced technologies, including artificial intelligence (AI), robotics, and other digital technologies. To continue its leadership and to satisfy the demands for large-scale compute, China has now become one of the biggest proponents and early-adopters of orbital data centres (ODCs).

In its approach, it has homed in on four elements: (1) top-down policy drafting and guidance, (2) private and state-owned industry adaptation, (3) municipal and provincial government incentivisation, and (4) supporting non-state actors and groups. Central to this effort have been the state-owned China Aerospace Science and Technology Corporation (CASC), who first formalised ODCs in a new strategy; the Beijing Municipal Government, and the Zhongguancun Science Park in Beijing; as well as a multitude of startups.

Timeline of China's ODC activities, 2025-2026



Source: Based on the authors' own compilation.

As China pursues orbital compute capabilities, the technological gap widens. Europe risks having to compute its (space) data through third-party actors as well as lacking access to orbital information. To ensure Europe's technology sovereignty is future-proof, this policy brief offers the following recommendations:

1. The EU should establish coordinated political guidance, investment support and regulatory framework for ODC developments.
2. The Horizon Europe Moonshot Projects under the 2028-2034 Multiannual Financial Framework (MFF) should include a flagship European ODC initiative.
3. ESA and EU should signal demand and offer associate contracts for in-space data processing for start-ups and scale-ups.
4. European actors should incentivise start-ups that address current feasibility challenges through a European ODC Challenge.
5. European space actors should convene policy and technology working groups with lead actors and companies worldwide on ODC developments and standards.

1 THE GLOBAL PUSH FOR ORBITAL DATA CENTRES: WHY NOW?

While European actors continue to weigh the development of orbital data centres (ODCs), Chinese institutions and companies are already moving towards implementation. China's Three-Body Constellation, launched by ADAspace in 2025, marked the first operational example of space-based computing; and Beijing has not slowed down since. In June 2026, the Beijing Municipal Science and Technology Commission and the Zhongguancun Science Park Management Committee issued a call for proposals for enabling technologies to support space-based computing satellites.

China's ambitions extend well beyond a single constellation. Beijing views space-based compute as the next frontier of its broader technology strategy: a way to produce advanced computing capabilities domestically and export them overseas—much like its U.S. counterparts—while positioning itself to help set the global standards and regulation for an entirely new industry before others get there first. **As China continues to expand its activities for ODCs at this pace, European actors should heed close attention or risk relinquishing space compute capabilities to the United States and China.**

Figure 1: Timeline of China's ODC activities, 2025-2026



Source: Based on the authors' own compilation.

This policy brief explores the strategic importance and role of ODCs, maps China's key developments and actors in the space that have arisen over the past year, analyses the next likely developments in China's space-compute industry, and makes policy recommendations for European stakeholders at the genesis of the next digital age.

1.1 Global demand and design of ODCs

ODCs or space-based data centres (SBDCs) are **a facility in space** that has **primary functions** linked to **data processing and storage**, a **computing payload** and **supporting**

subsystems and infrastructure.[1] There are three primary roles of ODCs: (1) **edge computing** at or near the source, (2) **fog or orchestration** for coordinating compute across multiple nodes and networks, and (3) **cloud compute** for general purpose computing and storage functions.

The way ODCs are designed and implemented influences the potential use scenarios and computing power available. ODC architectures range from **monolithic** systems (one large spacecraft or hosted platform carrying a substantial compute payload) to **distributed** systems, where compute is spread across many satellites and coordinated through inter-satellite links, optical communications, ground stations and terrestrial cloud infrastructure.

ODCs are a response to a real problem in the growth of space-generated data: Earth observation (EO) satellites, communications constellations, remote sensors, and non-terrestrial networks are producing ever-increasing volumes of data. Northern Sky Research forecasts a **14× increase in global space data volume by 2030**, equivalent to **500 exabytes** of data.[2] As seen in Figure 2, potential applications for ODCs are expected to emerge over the coming decade, with remote sensing having the most likely use case in the short-term.

Figure 2: Implementation timelines for ODC applications and use cases

Use case	Timeframe
Edge processing for remote sensing; Earth observation data reduction	Less than 5 years
Real-time AI inference (threat ID, security & intelligence)	5–10 years
In-constellation edge services; satellite autonomy	5–10 years
Defence and cyber applications	5–10 years
Data sovereignty and secure data vaults	More than 10 years
Remote and redundant storage (backup/cold)	5–10 years
Distributed global compute networks (constellations)	More than ten years
Large-scale AI training and long-horizon compute	More than ten years

Source: ESPI, Hein et al.

Centralising everything in ground-based cloud systems is becoming inefficient as bandwidth limits from transmitting raw satellite data to Earth are causing costly, delays, and operationally impractical consequences.[3] This creates the need for space-fairing

[1] Hein et al. Orbital Data Centres. Presentation to ESPI, 2026.
[2] GloveNewsWire. NSR Report: Space Traffic Data Volumes Increase 14x Over the Next Ten Years. NSR, 2021 (Link).
[3] This was stated by Li Chao, director of the space-based computing system research centre at Zhejiang Lab at the 2026 Space Day of China.

nations, including China, to overcome data-transport bottlenecks, optimise orbital data flow, minimise use of the ground network, and establish proximity to users.

1.2 China's rationale for pursuing ODCs in 2026

China has become one of the world's key players in the development of new advanced technologies, including artificial intelligence (AI), robotics, and other present-day digital technologies. To continue its leadership and to satisfy the demands for large-scale compute, China has now become one of the biggest proponents and early-adopters of the concept of ODCs.

While China's primary rationale in pursuing ODCs is increasing its compute capacity, it is also driven by global electricity and water constraints, which have already begun to surface across the world, such as in the United States.[4] Chinese startup Nayuta Space succinctly formulated the challenge, stating in a company announcement that "at both ends of the scale, the demand side for computing power is continuously increasing, but the supply side of computing power actually has theoretical physical limits" and that "in terms of electricity consumption, the International Energy Agency predicts that by 2030, global data centre electricity consumption will reach 945 terawatt-hours, surpassing Japan's current annual electricity consumption and accounting for nearly 3% of global electricity consumption."[5]

More broadly speaking, China's development of ODCs also falls within China's greater strategy of achieving **self-sufficiency in manufacturing and advanced technologies**, already outlined as early as 2015 in China's Made in China 2025 (中国制造2025) plan.[6] In contrast, Europeans may view this rationale for developing ODCs as pursuing digital sovereignty for "Europe's economic strength, security, and long-term competitiveness."[7]

These activities come at a time when China is increasingly betting on the emergence of homegrown AI companies and models as well as broader data storage and processing. In July 2026, China hosted its seventh version of the **annual World AI Conference (WAIC)** in Shanghai, where Chinese leader Xi Jinping gave the opening speech and China announced the establishment of the **World AI Cooperation Organisation (WAICO)**. In addition to this high-level policy momentum on AI, the WAIC also served as an exhibition forum for multiple space compute companies at the conference, outlined later in this brief and highlighting the increasing AI-space nexus.

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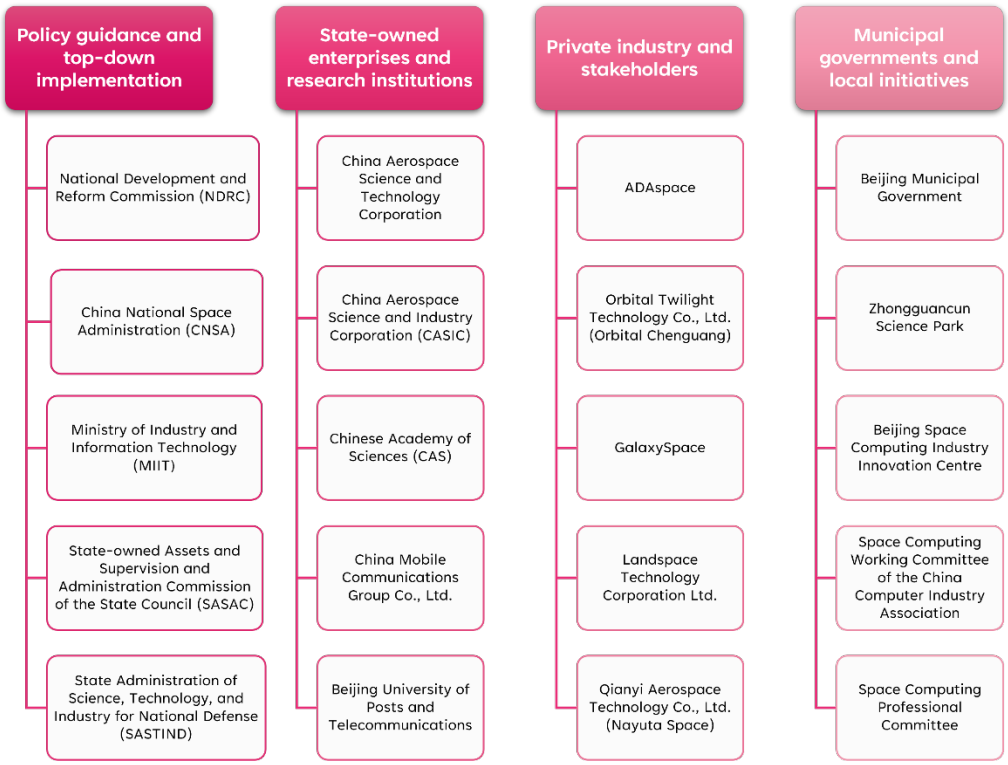
[4] European Space Policy Institute. Data centres in space: orbital backbone of the second digital era? ESPI, 2025 (Link).
[5] Nayuta Space. 国内首个星箭一体化GW级算力星座丨阿莱耶识ALAYA. Wechat, 2026 (Link); Jack Congram. Nayuta Space Exploring Using Rockets as Orbital Compute Platforms. China in Space, 2026 (Link).
[6] PRC State Council. Notice of the State Council on the Publication of "Made in China 2025." CSET, 2022 (Link).
[7] European Commission. Strengthening Europe's Tech Sovereignty. European Commission, 2026 (Link).

2 FROM THEORY TO PRACTICE: CHINA'S SPACE COMPUTE ADAPTATION

Over the past year, China's space ecosystem has turned towards ODCs as a solution to projected constraints. In contrast to privately-driven start-ups and early exploration in the U.S. and Europe, China has employed a whole-of-sector approach for ODCs, utilising policy guidance and technology implementation at once.

Specifically, China has made use of four elements: (1) top-down policy drafting and guidance, (2) private and state-owned industry adaptation, (3) municipal and provincial government incentivisation, and (4) supporting non-state actors and groups. Central to China's effort has been the state-owned CASC, who first formalised ODCs in a new strategy, the Beijing Municipal Government and the Zhongguancun Science Park in Beijing.

Figure 3: Key domestic stakeholders in China's emerging orbital datacentres industry



Source: based on authors' own data collection. Note: the table is not exhaustive and only maps key actors active on ODCs outlined in this brief.

2.1 China's policy guidance from the NDRC to CASC

Emerging domains like space have played an increasing role in China's state directives, recognised by the government as a key to global technological and political leadership in the future. Chinese government have increasingly turned towards drafting white papers and actions plans to help guide the space industry from the top-down.

Set against this backdrop, China's National Development and Reform Commission (NDRC) drafted its recommendations for its **15th five-year plan for the period 2026 to 2030** in October 2025, officially adopting them in March 2026 at China's 'Two Sessions.' The Chinese government recognised the **commercial space sector** (商业航天), **integrated circuits** (集成电路) and **embodied intelligence** (具身智能) as three out of ten key industries for the next five years.[8] While none of these policy recommendations name the potential use of ODCs yet, with these developments only entering early-stage development at the time these recommendations were drafted, it is likely that ODCs will officially be named in forthcoming space and AI policy documents, like China's forthcoming white paper on space expected in early 2027.

The adoption of ODCs in China is further highlighted by **CASC's five-year plan** in accordance with China's 15th five-year plan. In January 2026, CASC and the Chinese Society of Astronautics jointly organised the **Commercial Spacecraft and Application Industry Chain Joint Action Conference** (航天器及应用产业链共链行动大会) under the guidance of SASAC and CNSA, signalling high-level governmental support of the initiative. Participating organisations concluded three policy documents: the Vision Report on the Development of China's Commercial Space Industry During the 15th Five-Year Plan Period (十五五中国商业航天产业发展愿景报告), the Key Supply and Demand Information for China's Commercial Space Industry During the 15th Five-Year Plan Period ("十五五"中国商业航天重大供需信息), and the Standard System for the Commercial Spacecraft and Applications Industrial Chain During the 15th Five-Year Plan Period ("十五五"商业航天器及应用产业链标准体系). While it appears that no documents have been formally published, CGTN outlined that "plans include building gigawatt-level space digital infrastructure and creating a new integrated space system architecture combining cloud, edge, and terminal technologies" and stating that this "will enable deep integration of computing power, storage, and transmission capabilities, supporting space-based data processing and Earth-space collaborative computing."[9]

Additionally, the Chinese government already explicitly stated the need for data centre construction previously: in its **Global AI Governance Action Plan** in July 2025 by China's

[8] Embodied intelligence is "the most advanced form of artificial intelligence (AI) that develops through continuous physical interaction with the real world, tightly linking perception, learning and action within a physical body," according to an interview with Fuchun Sun, the vice dean of the Institute of Embodied Intelligence and Robotics at Tsinghua University in Beijing, published in Nature (Link).

[9] CGTN. China unveils 'space+' ambitions for tourism, mining and more. CGTN, 2026 (Link).

Ministry for Foreign Affairs, China states the need to construct "intelligent computing power, data centres and other infrastructure" and "promote the establishment of a unified computing power standard system."[10] Combined with other high-level policy documents, these plans give rise to move ODCs from theory to practice.

2.2 China's private sector: initial launch and implementation

China's industry has also begun to quickly move on developing ODCs: already in March 2025, the start-up ADAspace, a Chinese AI satellite company founded in 2018, launched the first group of its **Three-Body constellation**. As part of this constellation, ADAspace plans for this "'star compute' network [to] consist of 2,400 inference computing satellites and 400 training computing satellites, deployed in dawn-dusk orbits, sun-synchronous orbits and low-inclination orbits at altitudes between 500 and 1,000 kilometres."[11] While the next groups of satellites have not been launched yet, ADAspace has already completed initial tests, like tasking a robot from space with Shanghai Jiaotong University.[12]

New start-ups have also emerged to tackle the rising demand for space-based compute. In April 2026, **Beijing Orbital Twilight Technology Co., Ltd.** (Orbital Chenguang), registered in May 2025, announced raising an unknown amount of equity financing and €7.4 billion worth of debt financing.[13] **Nayuta Space** also announced in June 2026 its supercompute constellation ALAYA, planned to launch on Nayuta's own reusable rocket to "deploy 12,500 computing satellites into sun-synchronous orbit." At the WAIC in July 2026, **Shanghai Xingshu Tiansuan Space Technology Co.** also officially unveiled the launch of its 'Xingshu Plan,' Shanghai's own space-based computing envisioned to grow to 1,000 satellites.[14] Other companies and start-ups that have announced their own constellations include **China Mobile, Zhejiang Lab**, **Bailing Aerospace**, **Zhongke Tiansuan Technology**, and **Interstellar Glory Space-Based Computing Technology (iSpace)**.

2.3 Municipal and provincial initiatives for ODCs

Beyond the commercial sector, China's municipal governments and initiatives provide an incubating environment for new ODC companies. Beijing appears to have become the centre of this ODC industry, with Beijing's **Zhongguancun Technology and Innovation Park and the Beijing Municipal Science and Technology Commission** central to this effort and to broader technology initiatives. In June 2026, the **two institutions issued a call for proposals for ODC projects**, with each project eligible for a one-year funding period worth

[10] Ministry of Foreign Affairs of the People's Republic of China. Global AI Governance Action Plan. MFA, 2025 (Link).
[11] The State Council Information Office of the People's Republic of China. From ground to orbit: China eyes computing in space. SCIO, 2026 (Link).
[12] Ibid.
[13] Orbital Chenguang. 轨道辰光顺利完成新一轮股债权融资. Wechat, 2026 (Link).
[14] Yu, Cheng. Shanghai showcases satellites of its space-based computing network for AI. China Daily, 2026 (Link).

€644 thousand.[15] In this call, the two institutions prioritised the key technology areas of "four- or five-junction concentrator GaAs solar cells," "radiation-hardened 1,000 V silicon carbide" modules, and control and safe operation "technology for extra-large planar spacecraft." The call highlights China's ambition to build large-scale space infrastructure as well as tackle the core engineering challenges surrounding power and spacecraft scale.

In addition to funding, China also announced the launch of a **space computing innovation centre** in Beijing in June 2026. What makes this institute so unique is not its area of focus, but the participating institutions under the 'company+ alliance' model, such as the Beijing University of Posts and Telecommunications, Beijing Tiansuan Xinglian Technology Co., GalaxySpace, and Landspace—typically competitors in China's highly-competitive economy. According to CGTN, the centre's priority areas include "space-based computing chip design, high-performance computing payloads, satellite platforms and ground testing infrastructure, large model deployment, integrated space-ground cloud computing technologies, and satellite application innovation with scenario validation."[16]

2.4 Working groups and professional committees

As industry, research, and governmental resources coalesce around ODCs, China has also initiated two working groups dedicated to this endeavour. In April 2026, China established the first **'Space Computing Professional Committee**' on the sidelines of the 2026 Space Computing Industry Conference.[17] Hosted under the China Academy of Information and Communications Technology (CAICT), the professional committee will bring "together leading academics, industry pioneers, research institutions and financial organisations" and focus "on five core areas to systematically build a technological system and industrial ecosystem for space computing."[18]

Following up on this momentum, the **'Space Computing Working Committee of the China Computer Industry Association'** led under the chairmanship of Dr. Wang Jianyu at the Chinese Academy of Sciences (CAS), an expert on space-based optoelectronic technology and systems, was also established.[19] In line with the professional committee, the working committee was established to "systematically investigate the strategic layout and technological progress of space computing."[20]

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[15] Beijing International Centre for Science and Technology Innovation. 申报开启！2026年算力星座技术创新方向储备课题征集进行中. Wechat, 2026 (Link).
[16] CGTN. Beijing unveils space computing innovation centre to build full-stack computing ecosystem. CGTN, 2026 (Link).
[17] China in Space. China Moves to Manage Space-Based Computing Efforts. China in Space, 2026 (Link).
[18] Si, Ma. China's first Space Computing Professional Committee established. China Daily, 2026 (Link).
[19] Jones, Andrew. China builds institutional framework for space computing push. SpaceNews, 2026 (Link); Chinese Academy of Sciences. Wang Jianyu. CAS, n.d. (Link).
[20] Chinese Academy of Sciences. 太空计算工作委员会成立 业界呼吁加快形成太空计算生态-中新网. CAS, n.d. (Link).

3 CHINA'S BID FOR ODCS: IMPLICATIONS AND RECOMMENDATIONS FOR EUROPE

As this previous section lays out, China has adopted a top-down approach towards space computing, utilising local-to-central resources to shape and quickly tackle this new technology alongside China's industry.

3.1 Interpreting China's dual approach to ODCs

China's push for policy drafting at the same time as industry adoption however remains noteworthy, in comparison to the approaches adopted by the U.S. and Europe. Both actors have so far focused on industry adoption and early exploration, while China has steadily charged ahead with both. In the short-term, Chinese actors are likely to continue their bid for ODCs, leading to a rise in initiatives, incubators, and calls for proposals to foster new companies and approaches, as well as new top-level policy guidance.

As this brief outlines, there continue to be multiple constraints in deploying data centres. To abate some of these concerns, such as high water and electricity demand, the Chinese industry has also moved to explore **underwater datacentres**. In early 2026, the "world's first wind-powered underwater data centres (UDC)" was launched outside of Shanghai, featuring a "a total capacity of 24 MW."[21] The exploration of alternative data centre methods like UDCs are likely to continue alongside ODC innovation until this technology becomes more feasible.

3.2 A widening gap: consequences of China's ODC bid for Europe

China's investment in ODC technology and policy is bound to reverberate globally, setting standards, applications, and technologies for the global ODC sector in the future. Especially for European space actors who are pushing ahead on ODC exploration and implementation, China's activities will have profound impact, as terrestrial data centres are expected to hit limits in the future.

European space actors have also started to move on ODCs. Previously explored by ESPI, the European Union already funded the research project **ASCEND** in 2023 under Horizon Europe, the project found that ODCs "could strengthen European digital sovereignty." The European Space Agency (ESA) has also already implemented the Cognitive Cloud Computing in Space (3CS) together with a consortium led by Open Cosmos and launched the Φ-sat-2 satellite with AI capabilities. Multiple European companies have also moved to adopt ODCs, with French start-up ALATYR announcing plans in 2026 to manufacture

[21] China (Shanghai) Pilot Free Trade Zone Lin-Gang Special Area. World's first underwater data centre drives sustainable computing. Lingang Government, 2026 (Link).

robotically-assembled ODCs. Yet, as compared to the developments in China's space industry analysed in this brief and the U.S., the gap in Europe is starting to widen.

3.2.1 Policy recommendations: European steps for ODCs

As China, as well as the U.S., pursues orbital compute capabilities, Europe risks falling behind by inaction. If the technological gap widens significantly, Europe risks having to either compute its (space) data through third-party actors as well as lacking access to orbital information. This will prove crucial for terrestrial policy and decision-making in the future, especially as low-latency EO-based insights become the first crucial use case of ODCs.

Thus, Europe needs a sovereign response to ODCs or it will risk being beholden to the nations and companies that have put the appropriate architecture and standards in place. Looking at China's developments, this brief makes the following recommendations:

Figure 4: Policy recommendations on ODCs for European stakeholders

1. The EU should **recognise ODCs as an emerging pillar of European technological sovereignty** and establish coordinated political guidance, investment support and regulatory frameworks in support thereof.
2. As part of **Horizon Europe Moonshot Projects under the 2028-2034 MFF, a flagship European ODC initiative should be proposed.** In tandem, ESA should expand its CM28 programme proposals to include Computing beyond EO and Connectivity.
3. ESA and the EU, leveraging their respective programmatic and policy mandates, should adopt institutional proactivity by **signalling demand for space compute/in-space data processing** and **offering associate contracts** to established and upcoming companies.
4. European actors should **incentivise new approaches to current ODC feasibility challenges** and start a 'European ODC Challenge' to boost new initiatives.
5. European space actors **should convene policy and technology working groups** together with lead actors and companies worldwide on ODCs to establish policies, standards, and data sovereignty practices.

Looking towards China, European institutions and actors can learn and adapt from Beijing's playbook combining policy drafting with industry implementation for ODCs while navigating Europe's own political and economic constraints. In the short-term, Europeans can expect China's next white paper on space to be released in early 2027 to give more understanding of China's ODC development. Most importantly, China's early adoption of ODCs, combined with a changing geo-technological landscape also driven by other key actors like the U.S., should incentivise Europe to proactively respond to and shape space compute developments.



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